Asymmetric Siamese Networks for Semantic Change Detection in Aerial Images

Kunping Yang, Gui-Song Xia, *Senior Member, IEEE*, Zicheng Liu, Bo Du, *Senior Member, IEEE*, Wen Yang, *Senior Member, IEEE*, Marcello Pelillo, *Fellow, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Given two multitemporal aerial images, semantic change detection (SCD) aims to locate the land-cover variations and identify their change types with pixelwise boundaries. This problem is vital in many earth vision-related tasks, such as precise urban planning and natural resource management. Existing state-of-the-art algorithms mainly identify the changed pixels by applying homogeneous operations on each input image and comparing the extracted features. However, in changed regions, totally different land-cover distributions often require heterogeneous feature extraction procedures for images acquired at different times. In this article, we present an asymmetric Siamese network (ASN) to locate and identify semantic changes through feature pairs obtained from modules of widely different structures, which involves areas of various sizes and applies different quantities of parameters to factor in the discrepancy across land-cover distributions during different times. To better train and evaluate our model, we create a large-scale well-annotated SEmantic Change detectiON Dataset (SECOND), while an adaptive threshold learning (ATL) module and a separated kappa (SeK) coefficient are proposed to alleviate the influences of label imbalance in model training and evaluation. The experimental results demonstrate that the proposed model can stably outperform the state-of-the-art algorithms with different encoder backbones.**

***Index Terms*— Aerial images, asymmetric Siamese network (ASN), benchmark dataset, semantic change detection (SCD), separated kappa (SeK).**

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Kunping Yang, Zicheng Liu, and Liangpei Zhang are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan 430072, China (e-mail: kunpingyang@whu.edu.cn; zicheng.liu@whu.edu.cn; zlp62@whu.edu.cn).

Gui-Song Xia and Bo Du are with the National Engineering Research Center for Multimedia Software, the School of Computer Science, the Institute of Artificial Intelligence, and the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan 430072, China (e-mail: guisong.xia@whu.edu.cn; dubo@whu.edu.cn).

Wen Yang is with the School of Electronic Information, Wuhan University, Wuhan 430072, China (e-mail: yangwen@whu.edu.cn).

Marcello Pelillo is with the Department of Environmental Sciences, Informatics and Statistics (DAIS), Ca' Foscari University of Venice, 30172 Venezia, Italy (e-mail: pelillo@unive.it).

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I. Introduction

CHANGE detection in multitemporal aerial images [1]–[5], which aims to locate and analyze the regions of land-cover variations on the earth surface, is a crucial image interpretation task related to many applications, such as precise urban planning [6] and natural resource management [7]–[9]. Given a pair of multitemporal images, most existing methods focus on detecting the locations of changed pixels between the input images, namely, binary change detection (BCD), e.g., [1], [2], [10], [11]. However, since it overlooks pixels' categories, BCD often fails to depict the semantic change information that is highly demanded in subsequent applications. Hence, developing methods that can simultaneously extract changed regions and identify their land-cover classes in multitemporal images, i.e., semantic change detection (SCD), has become an active research topic in recent years [3], [12].

An intuitive solution to achieve SCD is to first partition the input images into semantic regions and then compare the segmentation results for identifying change types. However, this direct solution makes the underlying assumption that semantic categories in multitemporal images are independent and is typically troubled by two problematic aspects.

1) The changed regions of the same semantic category (such as building dismantlement and reconstruction for researching human activities and urban developments) cannot be distinguished.
2) The intrinsic correlation between semantic categories in images acquired at different times will be overlooked.

In order to take the advantage of categorical correlations in multitemporal images, many existing SCD methods rely on the architecture of Siamese networks [13] and have reported promising results [3], [12]. However, in contrast to the application scenarios in [14] and [15], the involved Siamese networks in SCD would face difficulties in locating and identifying changed regions conforming to different mixed land-cover distributions in images acquired at different times, which we call asymmetric changes. For instance in Fig. 1, when we identify the change types of land-cover changes, concentrating on a small area (framed out in orange rectangles) could provide delicate details to better identify regions with mixed distributions of trees and buildings around the marked positions (yellow points). However, to alleviate the ambiguity between buildings and impervious surfaces, image features involving more surroundings (framed out in red rectangles) are

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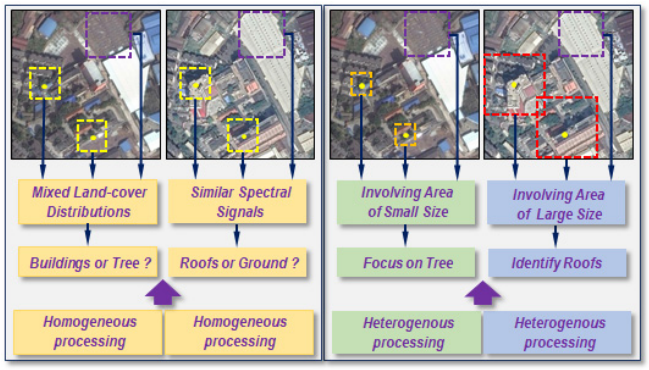


Fig. 1. In multitemporal aerial images, changed regions appearing at different spatial distributions during different times, which we call asymmetric changes, make it difficult to locate and analyze land-cover variations through existing methods with homogeneous image processings with respect to (w.r.t.) different image acquisition times. In contrast with existing methods, we are motivated to design some heterogeneous image processings, which we call local asymmetry, to factor in the discrepancy across different land-cover distributions and provide extra information for the SCD problem.

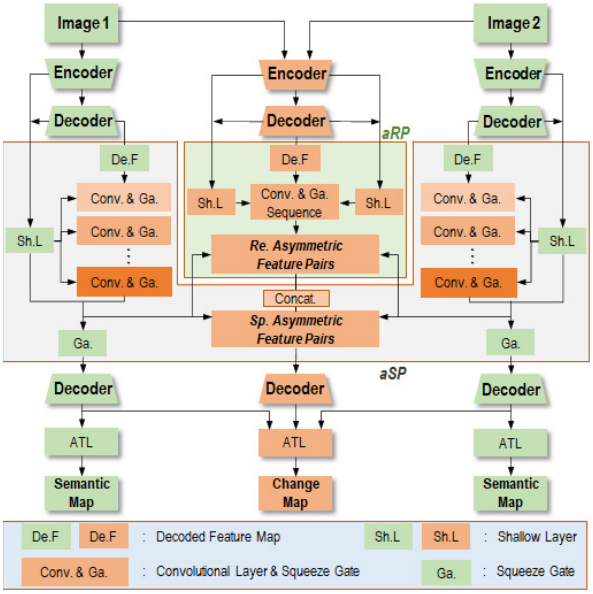


Fig. 2. Asymmetric Siamese network (ASN) for SCD. ASN utilizes Siamese encoders to map input multitemporal images into the feature space, while the Siamese decoders are leveraged to obtain semantic maps. Similarly, the encoder and decoder in the change detection branch are designed to obtain the change map. All the encoders and decoders for semantic labeling and change detection can be distinguished by colors. In contrast to the traditional Siamese network, ASN utilizes several different convolution sequences and squeeze gates in proposed asymmetric Spatial Pyramid (aSP) and asymmetric Representation Pyramid (aRP) to fuse feature pairs of different receptive fields and representation capabilities w.r.t. different image acquisition times, which we call asymmetric feature pairs, to provide extra information. Furthermore, the designed ATL is exploited to adaptively revise the output deflections based on the combinations of raw model outputs through slight extra convolution layers.

preferred. Furthermore, feature representation capabilities are often required to be adaptive. For example, pixels in one image belonging to objects with complex structures (e.g., buildings mixed with trees) often prefer modules with stronger representation capabilities, which, however, may make obvious flaws in those with simple structures (e.g., single impervious surface) in another image due to the overfitting. Besides, different objects sharing similar appearances sometimes need different feature representation capabilities (framed out in purple rectangles). In summary, various land-cover distributions in images acquired at different times cannot always be modeled well by solely using symmetric architectures as in Siamese networks. Thus, it is of great importance to design a deep model that can better depict the semantic but asymmetric changes in images.

In this article, we address the asymmetric properties of the SCD problem by exploiting Siamese networks. As illustrated in Fig. 2, we propose an asymmetric Siamese network (ASN) to extract changed pixels through two modules, i.e., asymmetric Spatial Pyramid (aSP) and asymmetric Representation Pyramid (aRP).

By leveraging designed feature-difference operations in aSP and aRP, multitemporal feature pairs independently deriving from different convolution sequences w.r.t. different image acquisition times (i.e., asymmetric feature pairs) are integrated. Consequently, the proposed aSP and aRP could focus on areas of different sizes through features of different representation capabilities w.r.t. different image acquisition times simultaneously during the change detection around one position, which we call the local asymmetry. By applying branch weights to emphasize adaptive convolution sequences, the proposed network can further be endowed with local asymmetries to adapt to different land-cover distributions during different times simultaneously, which is in contrast with traditional Siamese networks. As we shall see, ASN can better depict asymmetric changes between mixed objects and areas related to obscure land-cover categories.

To better train and evaluate the proposed model, we create a well-annotated SEmantic Change detectiON Dataset (SECOND) to set up a new benchmark. Although existing SCD datasets contain abundant categorical information, they are often not big enough, e.g., [12], which are inadequate to develop SCD algorithms with good generalization ability. Meanwhile, the annotations of some SCD datasets are unable to identify changes between the same land-cover class, e.g., [3]. The proposed SECOND is with 4662 pairs of images in 30 change types. Especially, we annotate the semantic categories and changed pixels separately in the SECOND dataset, which makes changed regions between the same land-cover class available.

Last but not least, model training and evaluation are always influenced by label imbalance due to overwhelming categories, e.g., nonchange pixels. Specifically, severe label imbalance would make models tend to collapse. Thus, we propose an adaptive threshold learning (ATL) module to adaptively revise the deflections of semantic change outputs caused by label imbalance through learnable output adjustments, where the logical relationships between each semantic category can be explored. Moreover, existing metrics used to evaluate change detection algorithms, such as overall accuracy (OA) and kappa coefficient (κ), are inherited from classification tasks, which would cause unreasonable scores due to the neglect of the dominant nonchange pixels. Thus, we further present a separated kappa (SeK) coefficient as a modified evaluation metric for the SCD task, which separates the nonchange class from other change types to reduce the effects of label imbalance.

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Compared with OA and κ, SeK is more in line with human scoring in the SCD problem.

Our main contributions in this article are threefold as follows.

1) We propose an asymmetric Siamese network, i.e., ASN, to factor in the discrepancy across different land-cover distributions in each multitemporal image, which can alleviate the categorical ambiguity through extra information provided by heterogeneous processings.
2) We create a large-scale semantic change detection dataset, i.e., SECOND, to better train deep models and as a new benchmark for the SCD problem. This SECOND also enables us to distinguish changed regions between the same land-cover class.
3) We design an adaptive threshold learning module and a separated kappa, i.e., ATL and SeK, to alleviate influences of label imbalances, which can adaptively revise the output deflections and fix unreasonable scores computed with traditional metrics, e.g., OA and κ, respectively.

II. Related Work

A. Localization of Changed Regions

In order to locate changed regions in multitemporal images, early works formulated the task as a BCD problem and relied on certain probability statistical formulations [1], [2], [16]–[19] or change vector analysis [20]–[24]. The statistical formulations and change vector analysis are able to explore the dissimilarity between features corresponding to changed and unchanged regions. Meanwhile, in order to embed pixelwise correspondences, graphical model-based algorithms are proposed [25], [26], which can model the spatial regularity during the optimization process. Further to expand the model capacities, deep learning networks [5], [27]–[30] are designed with elaborate structures to depict more diverse scenes by searching optimal parameters, where huge parameter spaces ensure stronger model capacities. Moreover, as illustrated in Section I, considering the discrepancy across land-cover distributions in input images could provide extra information when depicting land-cover distributions in the cases shown in Fig. 1. Thus, it is necessary to explore some asymmetric architecture in conventional deep networks. In this article, we propose an ASN to leverage locally asymmetric structures, which would be able to detect changed regions through the change detection branch more precisely compared with traditional Siamese networks.

B. Identification of Change Types

Existing SCD algorithms mainly utilize two ways to identify the change type of each pixel. The first kind of model, e.g., [12], consider nonchange as a special change type, which extracts unchanged regions and other change types simultaneously. The second kind of methods, e.g., [3], [31], separately extracts unchanged regions and identifies change types. However, these models often fail to consider issues, such as multiscale objects or scenes of various complexities in the interpretation process. Recently, increasing numbers of studies are focusing on enhancing deep features when capturing semantic labeling [32]–[34], among which parallel structures [35]–[37] w.r.t. diverse receptive fields significantly improve the model performances when dealing with multiscale objects. In view of this insight, in the SCD problem, we need to explore adaptive structures to obtain features suitable for different land-cover distributions corresponding to each input image, which motivates us to design parallel convolution sequences of different structures in ASN. Furthermore, dominant nonchange pixels and imbalanced land-cover category distributions in SCD problem would make models tend to collapse during the training process [38], which also leads us to investigate the revision process of raw model outputs. Many studies [39], [40] have been focusing on optimizing the model outputs through modeling the pixelwise correspondences, which causes repeated computations in the inference, such as the energy function optimization and recurrent neural module. To reduce the repeated computations and update semantic outputs together with change detection outputs simultaneously, we design a more efficient module, namely, ATL, which applies a few extra convolution layers on raw model outputs.

C. Datasets and Evaluation Metrics

Benchmark datasets and evaluation metrics are two important aspects related to training the designed models and measuring the generalization capability of SCD algorithms.

Existing datasets, e.g., [41]–[44], are mainly created for the BCD problem. The lack of land-cover categorical information still limits their usage in SCD although [5] has made an impressive contribution to alleviate data scarcity through active learning. A few benchmark datasets have been built for SCD [3], [12], [31]. Among them, the SCD datasets used in [12] and [31] are not big enough to sufficiently train and evaluate SCD algorithms, while the dataset in [3] utilizes two independently annotated land-cover maps to represent the change types, which ignores changed regions between the same land-cover class. Thus, in our proposed SECOND dataset, we check multitemporal images simultaneously to annotate land-cover classes and changed regions separately, which can distinguish changed regions between the same land-cover classes.

On the other hand, commonly used evaluation metrics for change detection tasks are mainly inherited from those used in classification problems, such as OA and κ used in [3] and [12], which ignores the fact that unchanged regions are often of the overwhelming majority in change maps. As a consequence, models with dominant predictions of nonchange pixels would get unreasonable high scores in terms of OA and κ. Thus, it is demanded to take into account the label imbalance for better evaluations. In this article, we utilize mean intersection over union (mIOU) [45] and design a new metric, i.e., SeK, to measure the SCD results, which can alleviate the label imbalance effects.

III. Methods

A. Problem Definition

Let $I_1, I_2 : \Omega \to \mathbb{R}^d$ denote two multitemporal images of $d \in \mathbb{N}_+$ channels, with Ω being the image grid $\{0, 1, \ldots, H-1\} \times \{0, 1, \ldots, W-1\}$. Given a set $L = \{y_1, \ldots, y_N\}$ of N

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semantic categories, the SCD problem aims to find a mapping function $f_{I_1,I_2} : \Omega \mapsto L^2 \cup \{(0,\,0)\}$ such that

$$\forall \mathbf{p} \in \Omega,\ \ f_{I_1,I_2}(\mathbf{p}) = \begin{cases} (0,\,0) & \text{if } \mathcal{C}_{I_1,I_2}(\mathbf{p}) < \tau, \\ (l_1,\,l_2), & \text{otherwise} \end{cases} \tag{1}$$

where $\mathcal{C}_{I_1,I_2}(\mathbf{p})$ measures the change probability of each pixel $\mathbf{p} \in \Omega$, $l_1, l_2 \in L$, and $(0,\,0)$ indicates the nonchange class. τ is a scalar thresholding on $\mathcal{C}_{I_1,I_2}$. Thus, f_{I_1,I_2} can locate changed regions and identify their categories simultaneously.

B. Intuitive Solution

An intuitive solution to obtain such a f_{I_1,I_2} is to first partition the input images I_1 and I_2 into semantic regions, e.g., with semantic segmentation algorithms, and then compare them to locate and identify semantic changes. Specifically, for $t = 1, 2$, let $\mathcal{M}_t : \Omega \mapsto \mathbb{R}^N$ be the semantic probability map of I_t, i.e., the probability vector $\mathcal{M}_t(\mathbf{p}) \in \mathbb{R}^N$ indicates the possibility of pixel $\mathbf{p} \in \Omega$ belonging to each semantic category in L. Denoting $l_t = \arg\max_{l \in L} \mathcal{M}_t(\mathbf{p})$ as the semantic category of pixel $\mathbf{p}$ in I_t, we have

$$f_{I_1,I_2}(\mathbf{p}) = \begin{cases} (0,\,0), & \text{if } l_1 = l_2 \\ (l_1,\,l_2), & \text{otherwise.} \end{cases} \tag{2}$$

As $\max \mathcal{M}_1(\mathbf{p}) \cdot \max \mathcal{M}_2(\mathbf{p}) = \max(\mathcal{M}_1^T(\mathbf{p}) \times \mathcal{M}_2(\mathbf{p}))$, for (2), we have

$$(l_1, l_2) = \arg \max_{(l_1,l_2) \in L^2} \left({\mathcal{M}_1}^T(\mathbf{p}) \times \mathcal{M}_2(\mathbf{p}) \right)$$

where the operation $\times$ represents the matrix product. However, due to the nonrelevance between $\mathcal{M}_1$ and $\mathcal{M}_2$, the intuitive solution actually implies that the semantic categories in each image are independent for every pixel. However, this underlying assumption is quite different from reality. The intrinsic correlation between each category is important, and the model performance would be limited by overlooking the categorical correlation. Moreover, according to (2), this intuitive solution also cannot identify changed regions between the same land-cover classes due to the same output form with nonchange pixels.

C. SCD With Conventional Siamese Networks

To take the categorical correlation into account, the state-of-the-art algorithm, i.e., HRSCD.str4 [3], utilizes Siamese semantic segmentation branches with an extra change detection branch to address the SCD problem. During the parameter optimization process, the Siamese branches would influence each other through the gradient flows and skip connections.

However, it is worth noticing that HRSCD.str4 relies on totally symmetric structures. More precisely, given a pair of multitemporal images, each response value in the feature map of the change detection branch is calculated based on the areas of the same size through nearly the same mapping function on the image acquired at different times. Besides, the Siamese decoders utilize single-line structures with skip connections, which are not fully aware of different object scale distributions in the image acquired at different times. As we shall see in Section VI, this symmetric architecture makes it difficult to locate and identify semantic changes related to widely different land-cover distributions across multitemporal images in some asymmetric changes.

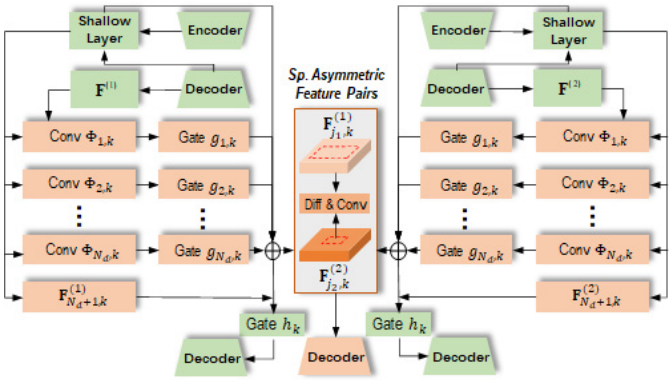


Fig. 3. Proposed aSP module with length of 1. By utilizing several Siamese convolution sequences $\Phi_{j,k}$ and squeeze gates $g_{j,k}$, decoded feature maps $\mathbf{F}^{(1)}, \mathbf{F}^{(2)}$ are employed to generate feature pairs $\mathbf{F}^{(1)}_{j_1,k}$ and $\mathbf{F}^{(2)}_{j_2,k}$ of different receptive fields w.r.t. different image acquisition times, while shallow layers are utilized to obtain branch weights and emphasize adaptive convolution sequences. Index k controls the channel numbers of layers in each convolution sequence, while j_1 and j_2 indicate different receptive fields. Each squeeze gate consists of the concatenation operator, convolution layers, and skip connections. Through this way, aSP can offer spatial (Sp.) asymmetric feature pairs of diverse spatial information to decoders for semantic labeling and change detection.

D. Asymmetric Siamese Network for SCD

1) Overall Architecture: In order to depict the aforementioned asymmetric changes that are beyond the descriptive capability of conventional Siamese networks, we propose to integrate features deriving from heterogeneous processes that are adaptive to each input. Following [3], we design the ASN with two Siamese branches for semantic segmentation and one branch for change detection (total three parallel branches). The basic architecture of each branch contains ten basic blocks (e.g., residual block) in the encoder and 13 basic blocks in the decoder, while five resize blocks are inserted following every two basic blocks beginning with the first basic block of encoder and decoder in each branch respectively (e.g., basic-basic-resize-$\cdots$-basic-basic-resize-$\cdots$). Specifically, as illustrated in Fig. 2, the proposed ASN exploits decoded Siamese feature maps, denoted as $\mathbf{F}^{(1)}, \mathbf{F}^{(2)}$ in Fig. 3, from the third resize block of the decoder in Siamese segmentation branches, while the decoded feature map, denoted as $\mathbf{F}^c$ in Fig. 4, from the third resize block of the decoder in change detection branch is also employed. As shown in Figs. 3 and 4, aSP obtains feature pairs $\mathbf{F}^{(1)}_{j_1,k}$, $\mathbf{F}^{(2)}_{j_2,k}$ based on $\mathbf{F}^{(1)}, \mathbf{F}^{(2)}$ (Siamese when given $j_1 = j_2$), while aRP further integrates $\mathbf{F}^{(1)}_{j_1,k}$, $\mathbf{F}^{(2)}_{j_2,k}$ with $\mathbf{F}^c$ to generates feature pairs $\mathbf{M}^{(1)}_{k_1}$, $\mathbf{M}^{(2)}_{k_2}$ (Siamese when given $k_1 = k_2$).

In designed ASN, we enumerate all possible combinations of $\mathbf{F}^{(1)}_{j_1,k}$, $\mathbf{F}^{(2)}_{j_2,k}$ and $\mathbf{M}^{(1)}_{k_1}$, $\mathbf{M}^{(2)}_{k_2}$ in pairs, respectively (e.g., $\mathbf{M}^{(1)}_{k_1}$ - $\mathbf{M}^{(2)}_{k_2}$ for a different k_1 and k_2), which independently derives from various convolution sequences w.r.t. different image acquisition times followed with squeeze gates to fuse

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the features and control the output channel numbers. With fewer output feature channel numbers, the computation costs caused by the following structures would be reduced. Consequently, the proposed ASN could focus on areas of different sizes through features of different representation capabilities w.r.t. different image acquisition times simultaneously during the change detection around one position. Meanwhile, similar to [46], we obtain branch weights based on shallow layers deriving from the third resize block in the encoder and decoder of each branch, where the global pooling and convolutions are applied, to emphasize adaptive convolution sequences. Afterward, $\mathbf{F}^{(1)}_{j_1,k}$ and $\mathbf{F}^{(2)}_{j_2,k}$ are integrated separately into Siamese decoders for semantic labeling, while all possible combinations of these feature pairs are integrated into the decoder for change detection. Finally, the proposed ATL adaptively revises the raw results of semantic labeling and change detection to compose the final SCD results.

2) Asymmetric Spatial Pyramid (aSP): In aSP, we design several parallel convolution sequences $\{\Phi_{j,k}\}_{1\le j\le N_d,\,1\le k\le N_r}$ of diverse structures, where each sequence is a convolutional operation set, denoted as $\Phi_{j,k}=\{\phi_{i,j,k}\}_{1\le i\le N_c}$. Specifically, N_d, N_c and N_r are positive integers, denoted as $\mathbb{N}_+$. For each $\phi_{i,j,k}$, the output channel number is set as $c_i\cdot r_k$, where channel hyperparameter $c_i\in\mathbb{N}_+$ and multiplication hyperparameter $r_k\in\mathbb{N}_+$ are used to control the feature representation capabilities. Besides, spatial hyperparameter $d_j\in\mathbb{N}_+$ is utilized to embed $\Phi_{j,k}$ with various spatial information when the dilation rate of $\phi_{i,j,k}$ is set as d_j.

As illustrated in Fig. 3, given decoded Siamese features $\mathbf{F}^{(1)},\mathbf{F}^{(2)}$ deriving from I_1 and I_2, respectively, in Siamese semantic segmentation branches, aSP integrates features calculated by each $\phi_{i,j,k}\in\Phi_{j,k}$. Then, we utilize squeeze gate $g_{j,k}$ consisting of convolutions and skip connections to fuse the features and reduce the computational complexity by controlling the channel numbers. Concretely, for $t=1,2$, we have

$$\mathbf{F}^{(t)}_{i,j,k}=v^{(t)}_{i,j}\cdot\phi_{i,j,k}\left(\mathbf{F}^{(t)}\right) \tag{3}$$

$$\mathbf{F}^{(t)}_{j,k}=g_{j,k}\left(\cup_{i=1}^{N_c}\mathbf{F}^{(t)}_{i,j,k},\,\mathbf{F}^{(t)}\right) \tag{4}$$

where $\cup$ is the concatenation operator. $v^{(t)}_{i,j}$ is the element in normalized branch weights $v^{(t)}\in\mathbb{R}^{N_c\cdot N_d}$, which are obtained through shallow layers deriving from the decoders of semantic segmentation branches. As a kind of attention weights, $v^{(t)}$ could be adjusted adaptively based on each input. The skip connection in squeeze gate $g_{j,k}$ connects $\mathbf{F}^{(t)}$ with the output to avoid gradient vanishing. The integrated feature map $\mathbf{F}^{(t)}_{j,k}$ embeds spatial information of various area sizes w.r.t. each index j for each time phase t. Given each index k, $\{\mathbf{F}^{(1)}_{j,k}\}_{1\le j\le N_d}$ and $\{\mathbf{F}^{(2)}_{j,k}\}_{1\le j\le N_d}$ compose a pair of Siamese spatial feature pyramid as in atrous spatial pyramid pooling (ASPP) [36].

Further to make receptive fields flexible and adaptive to each input during the changed region extraction, we design dense connected architectures to calculate the difference map between feature pairs $\mathbf{F}^{(1)}_{j,k}$ and $\mathbf{F}^{(2)}_{j,k}$ across Siamese branches through feature-difference operations. Given index k, for $1\le j_1,\,j_2\le N_d$, we have

$$\mathbf{M}_{j_1,j_2,k}=w^{(1)}_k\cdot\mathbf{F}^{(1)}_{j_1,k}-w^{(2)}_k\cdot\mathbf{F}^{(2)}_{j_2,k} \tag{5}$$

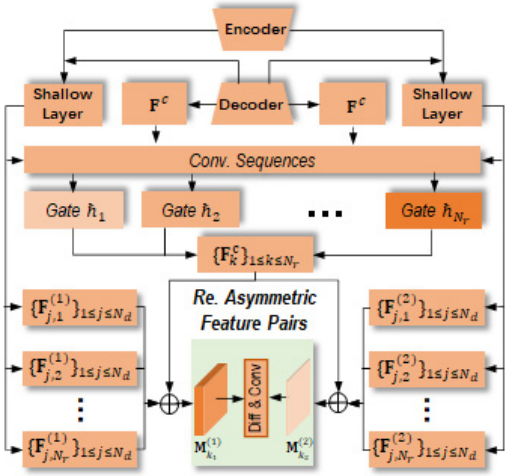


Fig. 4. By utilizing several Siamese convolution sequences and squeeze gates, aRP integrates the decoded feature map $\mathbf{F}^c$ with $\mathbf{F}^{(1)}_{j_1,k}$ and $\mathbf{F}^{(2)}_{j_2,k}$ to generate feature pairs $\mathbf{M}^{(1)}_{k_1}$ and $\mathbf{M}^{(2)}_{k_2}$ of different representation capabilities w.r.t. different image acquisition times. Also, shallow layers are utilized to obtain branch weights. Indices k_1 and k_2 indicate different feature representation capabilities. Each squeeze gate consists of the concatenation operator, convolution layers, and skip connections. Through this way, aRP can offer representation (Re.) asymmetric feature pairs of different representation capabilities to the decoder for change detection.

where $w^{(1)}_k,\,w^{(2)}_k\in\mathbb{R}^{N_r}$ are normalized branch weights calculated based on shallow layers deriving from the encoders of semantic segmentation branches. Feature pairs $\mathbf{F}^{(1)}_{j_1,k}$ and $\mathbf{F}^{(2)}_{j_2,k}$ are mostly generated independently of different structures and involve different sizes of image areas w.r.t. different image acquisition times, which makes the difference map $\mathbf{M}_{j_1,j_2,k}$ between $\mathbf{F}^{(1)}_{j_1,k}$ and $\mathbf{F}^{(2)}_{j_2,k}$ locally asymmetric in terms of spatial information. After the convolution operations, $\{\mathbf{M}_{j_1,j_2,k}\}_{1\le k\le N_r}$ are passed through the decoder in change detection branch to calculate the change probability map $\mathcal{C}_{I_1,I_2}$, where N_r represents the length of aSP.

Besides, we fuse $\mathbf{F}^{(t)}_{j,k}$ along index j, saying that, for $t=1,2$,

$$\mathbf{F}^{(t)}_k=h_k\left(w^{(t)}_k\cdot\cup_{j=1}^{N_d+1}\mathbf{F}^{(t)}_{j,k},\,\mathbf{F}^{(t)}_s\right) \tag{6}$$

where h_k represents the squeeze gate integrating global features, denoted as $\mathbf{F}^{(t)}_{N_d+1,k}$, with $\{\mathbf{F}^{(t)}_{j,k}\}_{j<N_d+1}$. Also, h_k connects shallow features $\mathbf{F}^{(t)}_s$ deriving from the decoders of semantic segmentation branches to avoid gradient vanishing. In this way, $\{\mathbf{F}^{(t)}_k\}_{1\le k\le N_r}$ of various receptive fields are passed through the Siamese decoder in Siamese semantic segmentation branches to obtain semantic probability maps $\mathcal{M}_1$ and $\mathcal{M}_2$. Then, semantic prediction maps $\mathcal{L}_{\mathcal{M}1}$ and $\mathcal{L}_{\mathcal{M}2}$ can be obtained by selecting the semantic category with the largest probability w.r.t. each position.

3) Asymmetric Representation Pyramid (aRP): As illustrated in Fig. 4, given decoded features $\mathbf{F}^c$ in change detection branch, we leverage convolution sequences and squeeze gates to obtain $\mathbf{F}^c_{j,k}$ in the same way as that obtaining $\mathbf{F}^{(t)}_{j,k}$. Moreover, based on shallow layers deriving from the encoder of change detection branch, we apply normalized branch weights $w^c=(w^c_1,\ldots,w^c_{N_r})\in\mathbb{R}^{N_r}$ to integrate $\mathbf{F}^c_{j,k}$ along the index j.

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Concretely, we have

$$\mathbf{F}_k^c = \hbar_k\left(w_k^c \cdot \cup_{j=1}^{N_d+1}\mathbf{F}_{j,k}^c,\ \mathbf{F}_s^c\right) \tag{7}$$

where $\hbar_k$ is also designed squeeze gate consisting of convolution layers and skip connections. Similarly, $\hbar_k$ concatenates global features, denoted as $\mathbf{F}_{N_d+1,k}^c$, with $\{\mathbf{F}_{j,k}^c\}_{j<N_d+1}$, while $\hbar_k$ connects shallow features $\mathbf{F}_s^c$ deriving from the decoder of change detection branch.

As illustrated in Fig. 4, we integrate $\mathbf{F}_k^c$ with $\{\mathbf{F}_{j,k}^{(1)}\}_{1\le j\le N_d}$ and $\{\mathbf{F}_{j,k}^{(2)}\}_{1\le j\le N_d}$ w.r.t. each index k by concatenation and convolution to obtain $\mathbf{M}_k^{(1)}$ and $\mathbf{M}_k^{(2)}$, respectively. All these $\{\mathbf{M}_k^{(1)}\}_{1\le k\le N_r}$, $\{\mathbf{M}_k^{(2)}\}_{1\le k\le N_r}$ compose a pair of Siamese representation feature pyramid in aRP, which are aligned feature maps of different representation capabilities.

Further to make feature representation capabilities flexible and adaptive to each input, we also calculate the difference map between $\mathbf{M}_k^{(1)}, \mathbf{M}_k^{(2)}$ in pairs through feature-difference operations. For $1 \le k_1, k_2 \le N_r$, we have

$$\mathbf{M}_{k_1,k_2}^c = \mathbf{M}_{k_1}^{(1)} - \mathbf{M}_{k_2}^{(2)}. \tag{8}$$

$\mathbf{M}_{k_1}^{(1)}$ and $\mathbf{M}_{k_2}^{(2)}$ contain different representation capabilities w.r.t. the indices k_1 and k_2, which makes the difference map $\mathbf{M}_{k_1,k_2}^c$ implicates different representation capabilities w.r.t. different image acquisition times, namely, locally asymmetric in terms of representation capabilities. Finally, after convolution operations, $\{\mathbf{M}_{k_1,k_2}^c\}_{1\le k_1,k_2\le N_r}$ are also passed through the decoder in change detection branch to calculate the change probability map $\mathcal{C}_{I_1,I_2}$.

Given the semantic prediction maps $\mathcal{L}_{\mathcal{M}1}, \mathcal{L}_{\mathcal{M}2}$ and change probability map $\mathcal{C}_{I_1,I_2}$, we can formulate our model as follows:

$$\forall \mathbf{p},\ f_{I_1,I_2}(\mathbf{p}) = \begin{cases}(0,0), & \mathcal{C}_{I_1,I_2}(\mathbf{p}) < \tau \\ \left(\mathcal{L}_{\mathcal{M}1}(\mathbf{p}), \mathcal{L}_{\mathcal{M}2}(\mathbf{p})\right), & \text{otherwise.}\end{cases}$$

4) Loss Function: We utilize fully supervised learning to optimize parameters in the proposed ASN. Specifically, given the ground truth $\mathcal{G}(\mathbf{p}, I_1, I_2)$, which not only locates changed pixels but also indicates semantic categories, we can get ground truth for semantic probability maps ($\mathcal{L}_{\mathcal{G}1}$ and $\mathcal{L}_{\mathcal{G}2}$) and change probability map $\mathcal{L}_{\mathcal{G}_c}$. Then, we have

$$\mathcal{L} = \alpha\mathcal{E}\left(\mathcal{M}_1, \mathcal{L}_{\mathcal{G}1}\right) + \beta\mathcal{E}\left(\mathcal{M}_2, \mathcal{L}_{\mathcal{G}2}\right) + \mathcal{E}\left(\mathcal{C}_{I_1,I_2}, \mathcal{L}_{\mathcal{G}_c}\right) \tag{9}$$

where $\mathcal{E}$ is the cross entropy function. α and β are loss weights to adjust respective loss terms. Stochastic gradient descent (SGD) is then used to reduce the total loss and obtain the optimal parameters, through which we train all branches simultaneously.

5) Adaptive Threshold Learning: As illustrated in Fig. 5, after the traditional training process, we design an ATL module, namely ATL, to search optimal thresholds based on raw model outputs. Given each probability map before softmax, denoted as $\mathcal{M}_1^{\text{raw}}$, $\mathcal{M}_2^{\text{raw}}$, and $\mathcal{C}_{I_1,I_2}^{\text{raw}}$, for $t = 1, 2$, we have

$$\hat{\mathcal{M}}_t^{\text{raw}} = \mathcal{M}_t^{\text{raw}} + \gamma\,\psi_1\left(\mathcal{M}_t^{\text{raw}}\right) \tag{10}$$

$$\hat{\mathcal{C}}_{I_1,I_2} = s_{\max}\left(\mathcal{C}_{I_1,I_2}^{\text{raw}} + \gamma\,\psi_2\left(\cup_{t=1}^2\hat{\mathcal{M}}_t^{\text{raw}}\right)\right) \tag{11}$$

where ψ_1 and ψ_2 are both convolution layer series with length of 2 and $s_{\max}$ represents softmax function.

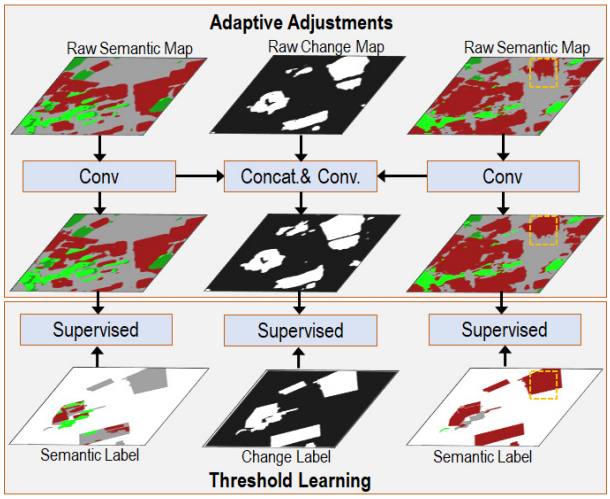


Fig. 5. Illustrations of the proposed ATL module. A few extra convolution layers are applied to optimize the raw semantic maps and change maps by modeling pixelwise correspondences. Without repeated computations in the inference, ATL causes fewer computation costs compared with classical postprocess, such as a conditional random field.

The hyperparameter γ is utilized to control the floating range when ATL adjusts the raw model outputs.

We fix the parameters in the whole model except ψ_1 and ψ_2 and retrain the model by utilizing the total loss in Section III-D4 calculated based on $s_{\max}(\hat{\mathcal{M}}_t^{\text{raw}})$ and $\hat{\mathcal{C}}_{I_1,I_2}$ with categorical weights. Thus, the model can be reformulated as

$$\hat{f}_{I_1,I_2}(\mathbf{p}) = \begin{cases}(0,0), & \hat{\mathcal{C}}_{I_1,I_2}(\mathbf{p}) < \tau \\ \left(\mathcal{L}_{\hat{\mathcal{M}}1}(\mathbf{p}), \mathcal{L}_{\hat{\mathcal{M}}2}(\mathbf{p})\right), & \text{otherwise.}\end{cases} \tag{12}$$

$\mathcal{L}_{\hat{\mathcal{M}}1}(\mathbf{p})$ and $\mathcal{L}_{\hat{\mathcal{M}}2}(\mathbf{p})$ are the updated semantic prediction maps obtained based on $s_{\max}(\hat{\mathcal{M}}_1^{\text{raw}})$ and $s_{\max}(\hat{\mathcal{M}}_2^{\text{raw}})$, respectively. During the parameter optimization, ATL explores the logical relationships between semantic categories within the considered neighborhood, which can adaptively update the semantic prediction maps and revise the threshold deflections caused by label imbalance.

IV. SECOND Dataset

Benchmark datasets are always essential elements for developing and testing data-driven aerial image understanding methods, e.g., [47]–[49]. Although several change detection datasets have been proposed [3], [41]–[44], [50], only a few of them contain land-cover categorical information, which is needed for SCD. A natural way to create a SCD dataset is comparing multitemporal land-cover maps in the same geographic locations [3], which would neglect change types, such as the demolition and reconstruction of buildings. Moreover, due to the required dense labors, a large-scale SCD dataset is hard to acquire, while the annotations of land-cover classes also require professional knowledge.

Following the principles in [48], in order to set up a new benchmark for SCD problems with adequate quantities, sufficient categories, and proper annotation methods, in this article, we present SECOND, a well-annotated SECOND dataset. Different from the datasets used in [12] and [31], to ensure

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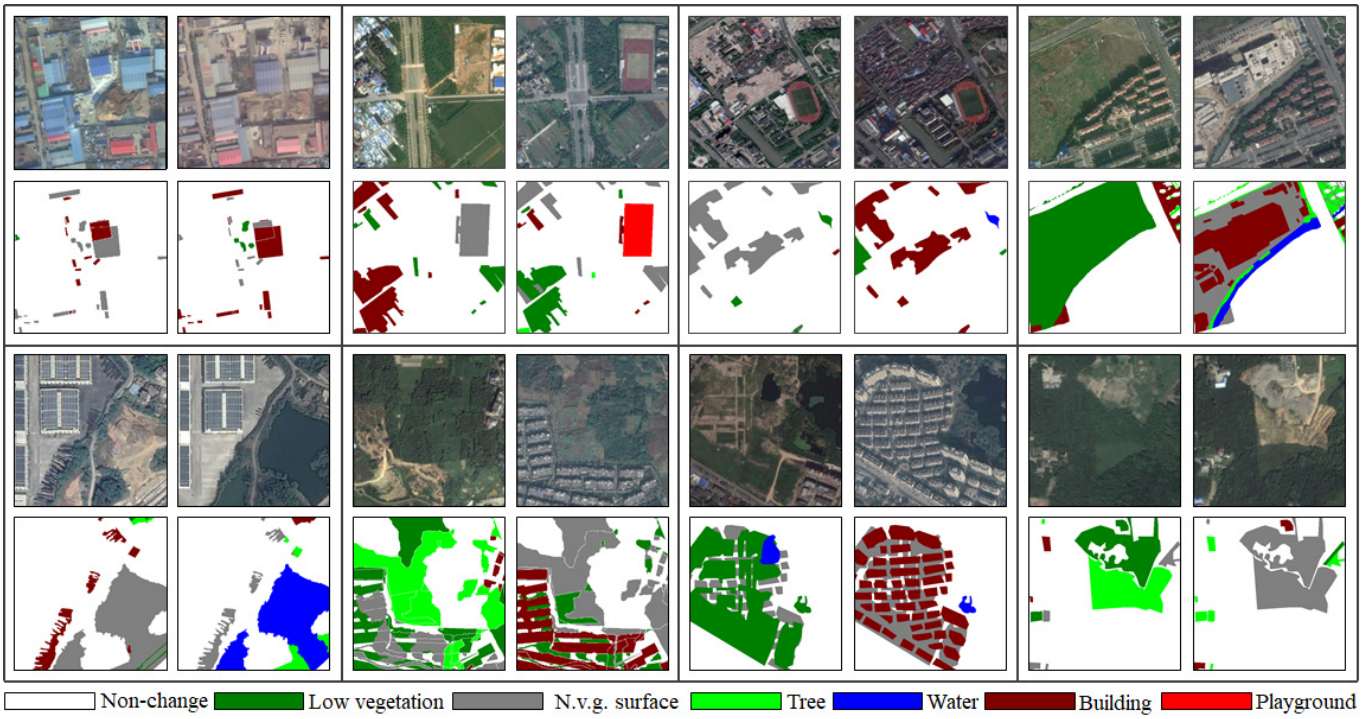


Fig. 6. Several samples of our proposed SECOND dataset. Color white indicates nonchange regions, while other colors indicate different land-cover classes. Ground truth for SCD can be obtained by comparing the annotated land-cover classes.

data diversity, we first collect 4662 pairs of aerial images from several platforms and sensors. These pairs of images are distributed over the cities, such as Hangzhou, Chengdu, and Shanghai. Each image has size 512 × 512 and is annotated at the pixel level. The annotation of SECOND is carried out by an expert group of earth vision applications, which guarantees high label accuracy.

Moreover, in contrast with [3], we utilize land-cover map pairs and nonchange masks to represent the change types. The nonchange masks equal the ground truth for BCD and are used to blacken the unchanged pixels in Siamese semantic segmentation branches during the training process, which can avoid semantic categorical ambiguity. Through this annotation method, we can distinguish nonchange pixels from changed pixels between the same land-cover class.

For the change type in the SECOND dataset, we focus on six main land-cover classes, i.e., nonvegetated ground surface, tree, low vegetation, water, buildings, and playgrounds, which are frequently involved in natural and man-made geographical changes [51]–[54]. It is worth noticing that, in the new dataset, nonvegetated ground surface (n.v.g. surface for short) mainly corresponds to the impervious surface and bare land. Due to the concerns on change types between the same land-cover category (e.g., changes between buildings), these six selected land-cover classes naturally raise 36 alternative change types. Therefore, 37 change types (including nonchange) are totally involved in. However, after the random selection of raw data samples, we find that some change types, such as changes between playgrounds and water, are nearly nonexistent in reality. Moreover, some other change types, such as changes between water, are impractical to define explicitly solely depending on visual characteristics. Consequently, we finally remain 30 change types in the SECOND, which can be found in Table II. Through the random selection of image pairs, the SECOND reflects real distributions of land-cover classes when changes occur. Several samples of SECOND are displayed in Fig. 6, where we can see the data diversity and label accuracy. The SECOND dataset and code would be available at https://captain-whu.github.io/SCD/.

V. Evaluation Metrics

A. Overall Accuracy and Kappa Coefficient

Existing works often utilize OA and Kappa coefficient (κ) that are commonly used for measuring classification performances to evaluate change detection algorithms [3], [12]. Given a confusion matrix $Q = \{q_{ij}\}$, where q_{ij} indicates the number of pixels that are identified as the ith change type and actually belong to the jth change type (nonchange is set as the first change type), then OA is defined as

$$\text{OA} \triangleq \rho = \sum_{i=1}^{C} q_{ii} \Big/ \sum_{i=1}^{C}\sum_{j=1}^{C} q_{ij} \tag{13}$$

where C is the total number of change types. Due to the equivalence of each pixel in the calculation of OA, dominant nonchange pixels would cause unreasonable scores, while, as a statistic calculated from confusion matrix (a kind of contingency table), κ measures the consistency between outputs and labels, which is less affected by the label imbalance [55]. More precisely, we have

$$\kappa = (\rho - \eta)/(1 - \eta) \tag{14}$$

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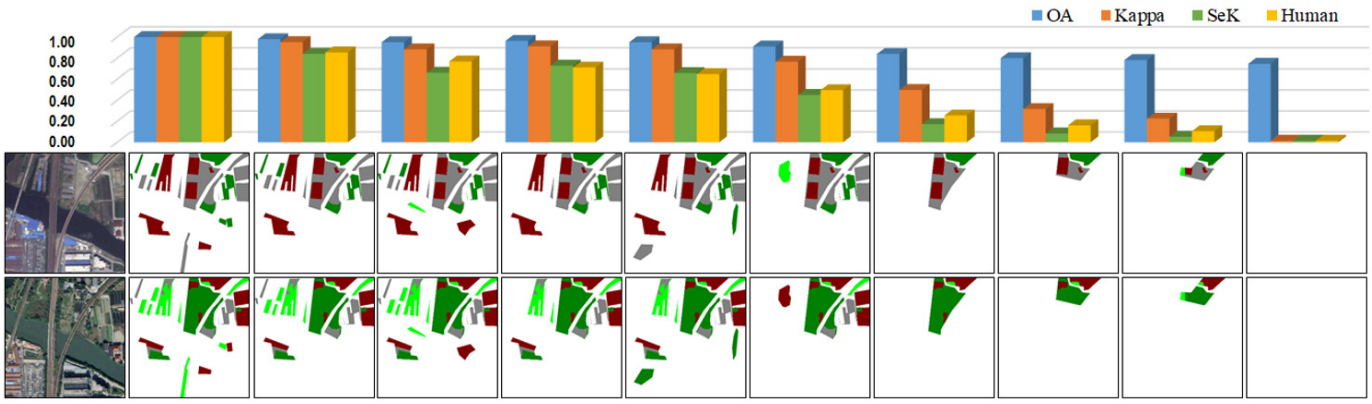


Fig. 7. Comparison between SeK and other metrics. The ground truth is listed in the leftmost. SeK alleviates the influence of label imbalance caused by nonchange pixels and shows most similarities with human scoring, while outputs with dominant predictions of nonchange get unreasonable high scores in OA and κ.

where $\eta = \sum_{j=1}^{C}(q_{j+} \cdot q_{+j})/(\sum_{i=1}^{C}\sum_{j=1}^{C} q_{ij})^2$ with q_{j+} and q_{+j} being as the row sum and column sum of the confusion matrix Q.

However, the dominant nonchange pixels still mislead the scores obtained by κ. Given a change detection data sample, i.e., a pair of images and a sequence of change detection results, we collect visual scores between 0 and 1 w.r.t. each result from 11 remote sensing image interpretation experts. Meanwhile, we calculate evaluation scores of each result based on OA and κ. As illustrated in Fig. 7, the collapse model with constant nonchange predictions would get unreasonable high scores in OA. Besides, although κ tends to zero when the model gradually collapses, the score of second to the last result still gets 0.23 in κ, which is too high compared with human scoring.

B. Separated Kappa (SeK) Coefficient

In order to alleviate the influence of label imbalance, we utilize mIOU to evaluate BCD results and propose an SeK coefficient to evaluate SCD results. Specifically, given a confusion matrix Q, we have

$$\mathrm{IOU}_1 = q_{11} \Big/ \left(\sum_{i=1}^{C} q_{i1} + \sum_{j=1}^{C} q_{1j} - q_{11} \right) \tag{15}$$

$$\mathrm{IOU}_2 = \sum_{i=2}^{C}\sum_{j=2}^{C} q_{ij} \Big/ \left(\sum_{i=1}^{C}\sum_{j=1}^{C} q_{ij} - q_{11} \right) \tag{16}$$

where IOU_1 measures the identification of nonchange pixels and IOU_2 evaluates the extraction of changed regions. Then, we have

$$\mathrm{mIOU} = \frac{1}{2}(\mathrm{IOU}_1 + \mathrm{IOU}_2). \tag{17}$$

Involving in IOU_2, mIOU considers more about changed regions.

On the other hand, the true positive of nonchange pixels q_{11} always dominates the calculation of κ. Thus, we separate q_{11} in the calculation of SeK. We also utilize IOU_2 to further emphasize changed pixels. Specifically, we define

$$\mathrm{SeK} = e^{(\mathrm{IOU}_2 - 1)} \cdot (\hat{\rho} - \hat{\eta})/(1 - \hat{\eta}) \tag{18}$$

with

$$\hat{\rho} = \sum_{i=2}^{C} q_{ii} \Big/ \left(\sum_{i=1}^{C}\sum_{j=1}^{C} q_{ij} - q_{11} \right)$$

$$\hat{\eta} = \sum_{j=1}^{C} (\hat{q}_{j+} \cdot \hat{q}_{+j}) \Big/ \left(\sum_{i=1}^{C}\sum_{j=1}^{C} q_{ij} - q_{11} \right)^2$$

where $\hat{q}_{j+}$ and $\hat{q}_{+j}$ represent the row sum and column sum of the confusion matrix without q_{11}. The exponential form enlarges the discernibility compared with simple multiplication when evaluating models with better performances.

As illustrated in Fig. 7, compared with κ and OA, models with apparently poor performances on small change types would get low scores in SeK no matter how good the performances on BCD are. Moreover, the mean square error (MSE) between SeK and human scores is 0.003. While, MSEs w.r.t. OA and κ are 0.212 and 0.028, respectively, which further validates the rationality of SeK.

VI. Experiments and Analysis

In this section, we evaluate the proposed ASN on the SECOND dataset. We first clarify the experiment settings in Section VI-A. Then, in Sections VI-B and VI-C, we discuss several existing structures and compare ASN with several methods using backbones with different basic blocks (e.g., [56]–[58]), including natural extensions of classical BCD algorithms and the state-of-the-art SCD algorithms [3], [27], to analyze the basic structures for SCD and verify the effects of proposed modules. Furthermore, the natural extensions of one latest BCD algorithm [59] are implemented to enhance the experimental results. Furthermore, in Section VI-D, we demonstrate the effectiveness of our proposed modules by feature visualization. Finally, in Section VI-E, we design the ablation study to verify the merits of each term in the proposed model.

A. Experiment Settings

The change detection algorithms involved in the comparison experiments are given as follows.

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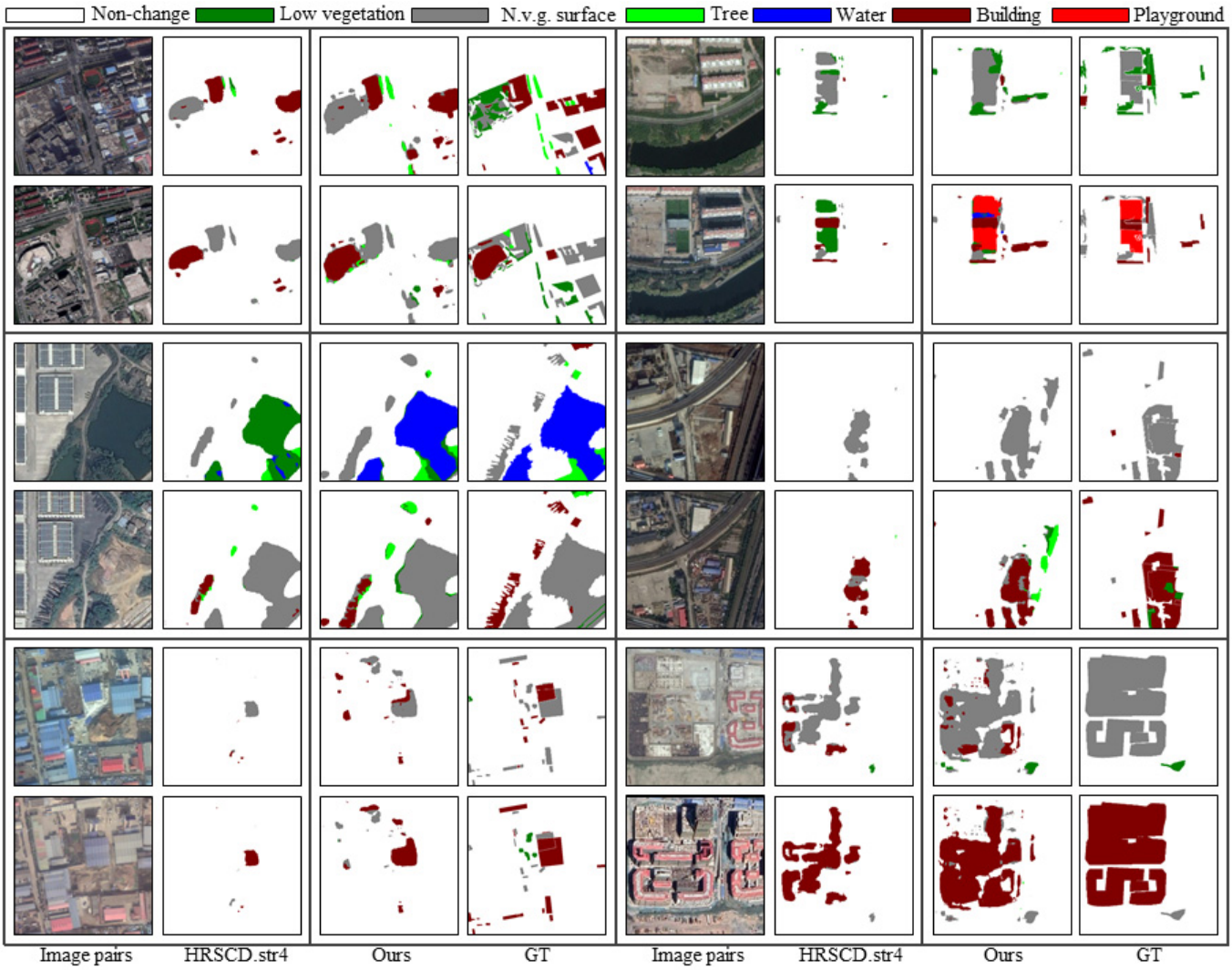


Fig. 8. Visual results of comparison with the state-of-the-art method when the encoder is built on residual blocks. We mask the semantic prediction maps with change maps to represent the prediction of change type in each position, where our proposed ASN can better identify land-cover classes and alleviate false identifications of nonchange pixels.

1) *FC-EF [27]:* A BCD algorithm using single encoder–decoder structure.
2) *FC-conc [27]:* A BCD algorithm using Siamese encoders followed with single decoder branch and concatenation skip connections from the encoder to the decoder.
3) *FC-diff [27]:* A BCD algorithm using Siamese encoders followed with single decoder branch and difference skip connections from the encoder to the decoder.
4) *DASNet [59]:* A BCD algorithm using dual-attention mechanisms.
5) *HRSCD.str1 [3]:* An algorithm corresponding to the intuitive solution to SCD problem discussed in Section III-B.
6) *HRSCD.str2 [3]:* An SCD algorithm using single encoder–decoder structure.
7) *HRSCD.str3 [3]:* An SCD algorithm using Siamese semantic segmentation branches with change detection branch.
8) *HRSCD.str4 [3]:* An SCD algorithm using Siamese semantic segmentation branches with change detection branch and difference skip connections from Siamese encoders to the decoder of change detection branch.

ASN keeps the same basic architecture as HRSCD.str4. We replace the difference skip connections with simple summations in ASN to reduce model sizes. ASN-ATL applies proposed adaptive threshold learning module on ASN. All the experiments are implemented on 4 Pascal V100 with a memory of 16 GB based on Pytorch. All evaluated models are trained and tested under the same conditions without pretrained models and other postprocessing.

In the training process, SGD is utilized to search optimal parameters for 50 epochs, while extra parameters in ASN-ATL would be trained for 20 epochs with other parameters fixed. Random flip and random scale between 0.5 and 2 are utilized as the data augmentation. The initial learning rate is set as 0.005, and the "poly" policy is with the power of 0.9.

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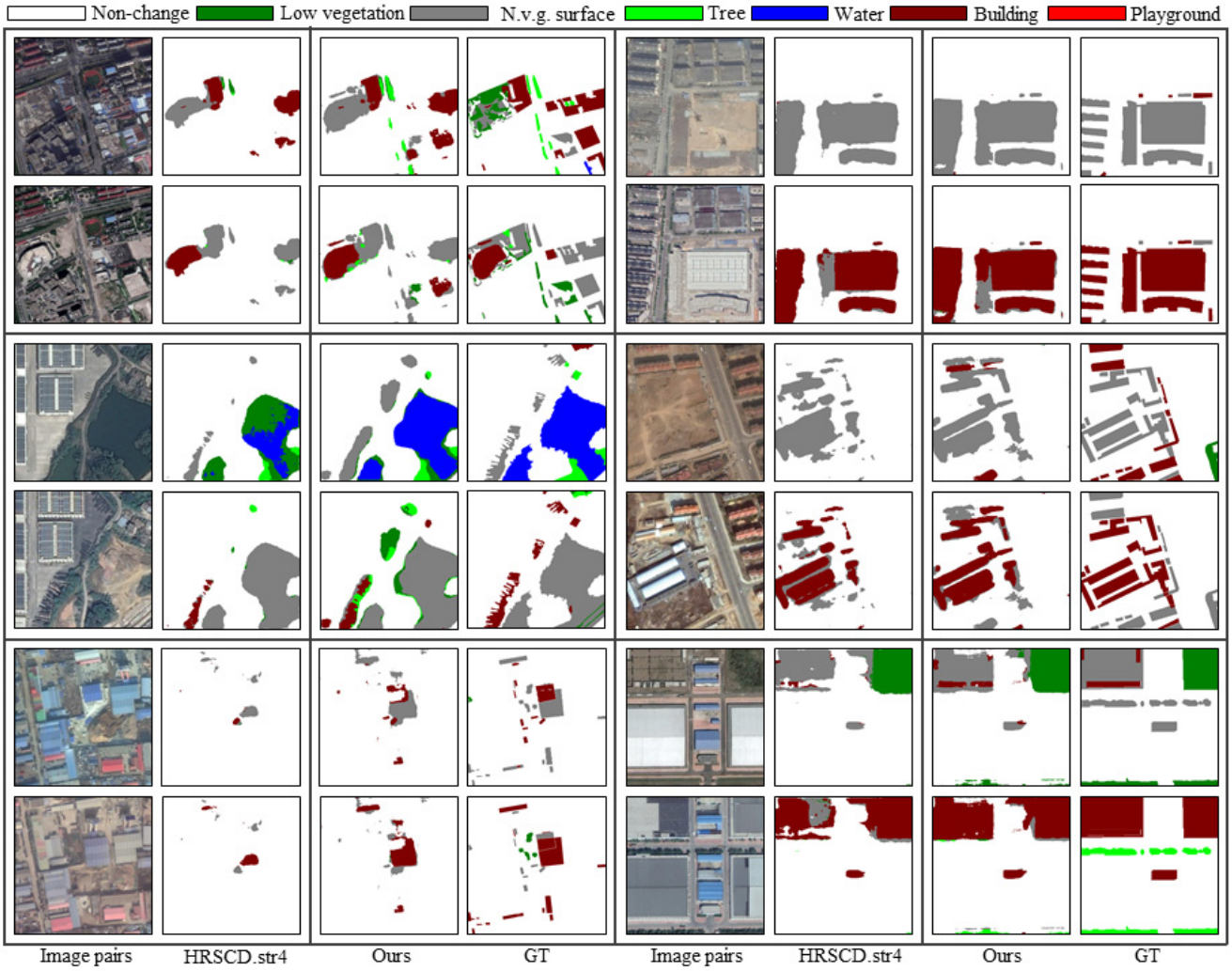


Fig. 9. Comparisons with the state-of-the-art method when the encoder is built on Xception blocks. Left part of the figure shows the stable outperformance of ASN with different backbones, while the right part exhibits the generalization ability on different data samples.

The batch size is set as 4. Also, the momentum is set as 0.9, and the weight decay is set as 0.0001. In the testing process, we apply flip strategy and multiscale (MS) testing with six scales that are 0.5, 0.75, 1.0, 1.25, 1.5, and 1.75.

As for the hyperparameters in ASN, considering the image size in SECOND, we set spatial hyperparameters $\{d_1, \ldots, d_{N_d}\}$ as $\{0, 6, 12\}$ to make feature receptive fields in aSP vary from local to the whole image. Besides, multiplication hyperparameters $\{r_1, \ldots, r_{N_r}\}$ and channel hyperparameters $\{c_1, \ldots, c_{N_c}\}$ are set as $\{16, 32, 64\}$ and $\{1, 2, 3, 4, 5\}$ to make features in aSP and aRP contain balanced representation capabilities with features in decoders. The kernel sizes of convolution layers in aSP and aRP are all set as 3. We further set the scalar threshold τ as 0.5. The hyperparameter γ in the ATL module is also set as 0.5, while we slightly adjust it to 0.4 for the Xception backbone. For all the experiments, we split the SECOND dataset into two subsets: 2968 sample pairs for training and 1694 sample pairs for testing. Finally, we set α and β in total loss as 1 in the training process.

TABLE I

COMPARISONS WITH STATE-OF-THE-ART METHODS WHEN THE ENCODER IS BUILT ON RESIDUAL BLOCKS

Methods	MS __ Flip __		MS ✓ Flip __		MS ✓ Flip ✓	
	mIOU	SeK	mIOU	SeK	mIOU	SeK
FC-EF [27]	59.3	5.7	59.3	5.7	59.0	5.9
FC-conc [27]	63.3	9.1	62.8	9.2	62.9	9.4
FC-diff [27]	61.9	8.8	60.9	8.6	61.0	8.7
DASNet [59]	57.1	6.5	56.1	6.4	55.5	6.2
HRSCD.str1 [3]	29.3	4.6	29.8	4.9	29.8	4.9
HRSCD.str2 [3]	59.7	6.3	59.4	6.5	59.4	6.6
HRSCD.str3 [3]	62.3	8.9	62.0	9.1	62.1	9.2
HRSCD.str4 [3]	67.5	13.7	67.8	14.4	67.9	14.5
ASN (Ours)	69.0	15.2	69.5	16.1	69.7	16.2
ASN-ATL (Ours)	**69.1**	**15.5**	**69.8**	**16.5**	**70.0**	**16.8**

B. Discussions on Basic Structures

As illustrated in Table I, HRSCD.str1 gets 29.8 in mIOU with full testing strategies, while FC-conc and FC-diff achieve

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TABLE II

DETAIL CATEGORICAL RESULTS WITH THREE KINDS OF TESTING STRATEGIES WHEN THE ENCODER IS BUILT ON RESIDUAL BLOCKS. TOP SHEETS ARE WITHOUT TESTING STRATEGY. MIDDLE SHEETS ARE WITH MS TESTING STRATEGY. BOTTOM SHEETS ARE WITH MS AND FLIP TESTING STRATEGY. THE CATEGORICAL SEK IS LISTED IN THE MATRICES, WHILE THE CATEGORICAL INTERSECTION OVER UNION (IOU_1, IOU_2) OF BINARY CHANGE DETECTION (BCD) IS LISTED BELOW EACH MATRIX

Residual	HRSCD.str1(intuitive solution)						HRSCD.str4						ASN-ATL (Ours)					
	Low vegetation	N.v.g. surface	Building	Tree	Water	Playground	Low vegetation	N.v.g. surface	Building	Tree	Water	Playground	Low vegetation	N.v.g. surface	Building	Tree	Water	Playground
Low vegetation	–	30.0	22.7	19.4	21.7	3.3	–	41.7	30.8	22.2	19.9	8.4	**–**	**42.0**	**32.2**	**28.3**	**20.4**	**34.2**
N.v.g. surface	31.3	–	31.3	20.7	15.0	4.7	**44.1**	–	42.1	28.5	24.6	6.0	43.7	–	**44.7**	**30.4**	**29.5**	**41.9**
Building	21.9	31.3	21.4	12.6	19.2	0.0	**35.1**	47.2	26.1	18.1	**33.8**	0.0	33.1	**48.3**	**27.1**	**24.9**	31.1	**17.8**
Tree	16.9	19.1	16.1	–	1.1	0.0	12.0	26.0	**24.5**	**–**	**12.6**	0.0	**20.2**	**27.7**	23.7	–	5.9	0.0
Water	19.5	16.1	19.0	10.0	–	–	13.7	26.0	**29.2**	13.0	–	–	**25.9**	**29.8**	27.8	**21.7**	–	–
Playground	0.1	3.4	0.0	0.0	–	–	1.1	1.8	0.4	0.0	–	–	**17.1**	**29.7**	**16.5**	**1.3**	–	–
Non-change	IOU_1 : 34.9			IOU_2 : 23.8			IOU_1 : **86.6**			IOU_2 : 48.4			IOU_1 :85.9			IOU_2 :**52.4**		
Low vegetation	–	30.6	23.1	20.0	**22.0**	1.2	–	42.4	31.6	21.8	14.4	4.7	**–**	**43.4**	**32.5**	**28.5**	21.2	**37.6**
N.v.g. surface	31.5	–	31.9	21.4	14.3	1.7	44.7	–	42.8	29.2	19.8	4.8	**45.4**	**–**	**45.4**	**31.5**	**31.5**	**43.3**
Building	22.4	32.3	22.5	13.9	20.5	0.0	**36.1**	48.3	27.9	21.4	31.2	0.0	34.4	**49.4**	**28.7**	**25.9**	**36.0**	**20.1**
Tree	17.8	19.8	16.9	–	0.9	0.0	12.0	26.8	**25.6**	–	7.8	0.0	**21.0**	**28.7**	24.8	–	**10.3**	0.0
Water	16.4	15.9	17.8	11.0	–	–	7.5	22.0	26.3	13.0	–	–	**26.8**	**31.4**	**28.8**	**21.3**	–	–
Playground	0.0	1.9	0.0	0.0	–	–	0.1	1.7	0.3	0.0	–	–	**15.5**	**28.0**	**20.8**	**0.1**	–	–
Non-change	IOU_1 : 35.6			IOU_2 : 24.0			IOU_1 : **86.9**			IOU_2 : 48.8			IOU_1 : 86.4			IOU_2 : **53.3**		
Low vegetation	–	30.8	23.2	20.1	**21.9**	1.2	–	42.6	31.8	22.2	11.3	3.5	**–**	**43.6**	**32.7**	**28.3**	21.5	**37.7**
N.v.g. surface	31.6	–	31.9	21.4	14.1	2.3	45.0	–	42.9	29.4	18.9	4.4	**45.7**	**–**	**45.7**	**31.8**	**32.5**	**43.3**
Building	22.5	32.4	22.4	14.2	20.5	0.0	**36.3**	48.4	28.2	22.0	29.3	0.0	34.6	**49.7**	**29.1**	**26.4**	**35.5**	**20.1**
Tree	18.0	20.0	16.8	–	0.8	0.0	12.8	26.7	**26.2**	–	6.4	0.0	**21.3**	**28.6**	25.2	–	**13.4**	0.0
Water	16.1	15.4	18.1	9.6	–	–	8.7	22.2	25.7	15.4	–	–	**27.4**	**32.1**	**29.2**	**22.0**	–	–
Playground	0.0	2.4	0.0	0.0	–	–	0.0	1.6	0.3	0.0	–	–	**15.9**	**27.9**	**24.9**	**0.3**	–	–
Non-change	IOU_1 : 35.6			IOU_2 : 24.0			IOU_1 : **87.0**			IOU_2 : 48.8			IOU_1 : 86.5			IOU_2 : **53.5**		

62.9 and 61.0 respectively. This result demonstrates that separate semantic segmentation application and false categorical independence assumption would limit the model performance, which could be addressed by the joint training of the Siamese network. Importantly, although FC-EF outperforms HRSCD.str1 dramatically in mIOU, they share similar results in SeK, which implicates that SeK would not simply increase merely with good BCD performance. The outperformances of FC-conc and FC-diff compared with FC-EF indicate that the Siamese networks provide extra improvements compared with single encoder–decoder structure. We can further conclude from Table I that SCD results cannot be simply improved with more elaborately designed structures for CD problems, such as those in DASNet. Moreover, the outperformances of HRSCD.str3 and HRSCD.str4 show the superiority of separate changed region extraction and change type identification.

C. Comparison With State-of-the-Art Algorithms

As shown in Table I, ASN-ATL achieves the best results, which are 69.1 in mIOU and 15.5 in SeK, without testing strategies. Then, with a multiscale strategy, ASN-ATL achieves 69.8 in mIOU and 16.5 in SeK, while the flip strategy provides extra 0.2 improvements in mIOU and 0.3 improvements in SeK. Our proposed model stably outperforms existing state-of-the-art algorithms, which achieves 2.1 improvements in mIOU and 2.3 improvements in SeK compared to HRSCD.str4 with full testing strategies. Also, as shown in Table I, ATL provides 0.3 improvements in mIOU and 0.6 improvements in SeK with full testing strategies. We can also see in Fig. 8 that the proposed model can recover more details, which leads to better performance in SeK.

To further explore the detail comparison, we list the categorical SeK (calculated from 2×2 confusion matrix concentrating on each change type) and categorical intersection over union (IOU_1 and IOU_2) in Table II. Each row (except nonchange) indicates land-cover classes in the first image, while each column indicates land-cover classes in the second image. Categorical SeK evaluates the identification of all change types except nonchange, while categorical intersection over union measures the results of nonchange pixel extraction. We choose HRSCD.str4, HRSCD.str1, and ASN-ATL to be listed in Table II for detail comparison. We can see that ASN-ATL achieves the best performance on an overwhelming majority of change types under all kinds of testing strategies. Furthermore, we can conclude from the third row and the third column of each matrix in Table II that the proposed model can better identify the change type between the same land-cover category (i.e., changes between buildings

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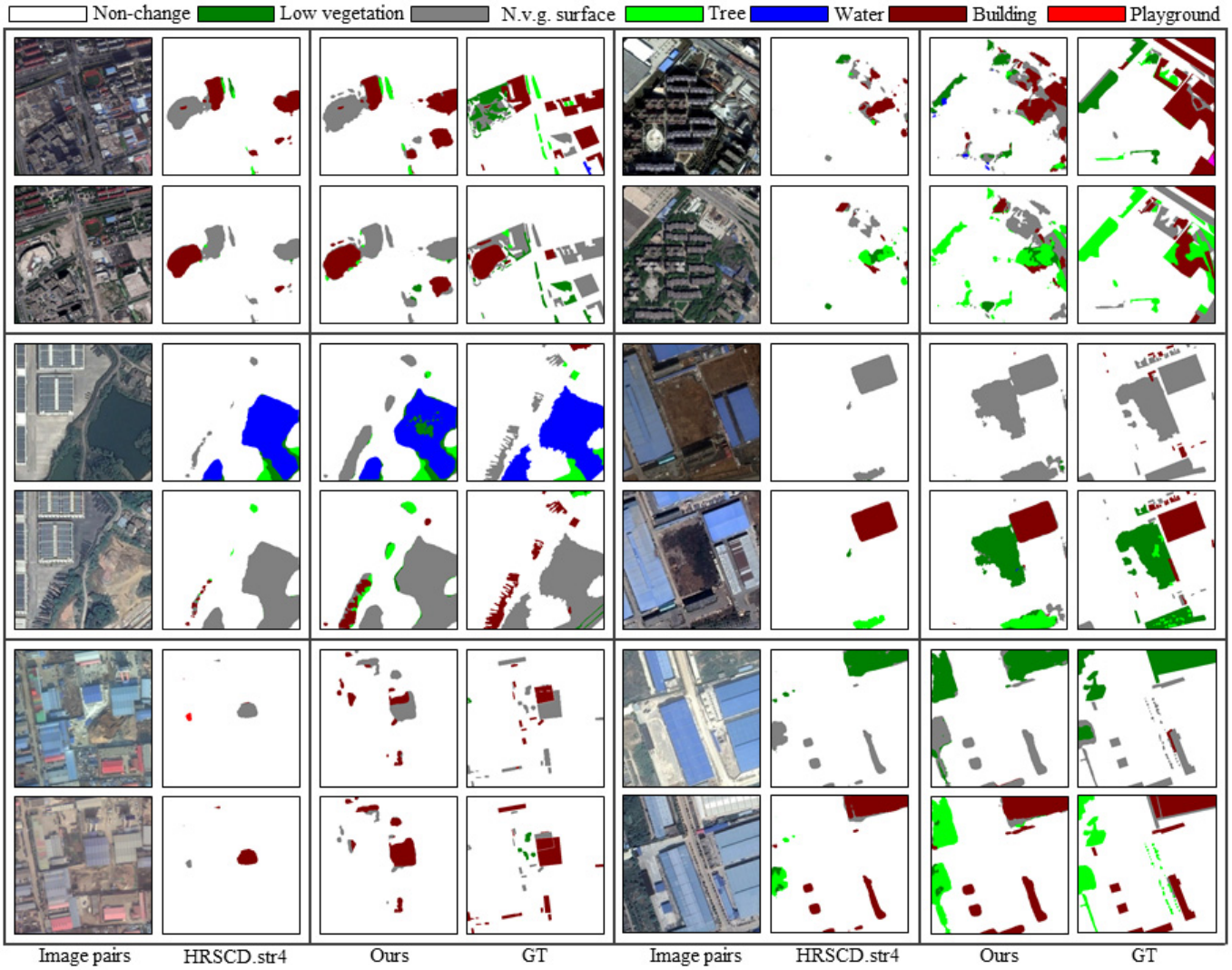


Fig. 10. Comparisons with the state-of-the-art method when the encoder is built on squeeze-and-excitation blocks. ASN also shows stable the outperformance and generalization ability.

TABLE III

COMPARISONS WITH STATE-OF-THE-ART METHODS WHEN THE ENCODER IS BUILT ON XCEPTION BLOCKS

Methods	MS __ Flip __		MS ✓ Flip __		MS ✓ Flip ✓	
	mIOU	SeK	mIOU	SeK	mIOU	SeK
FC-EF [27]	56.2	3.6	55.6	3.9	55.7	4.0
FC-conc [27]	61.1	7.3	60.4	7.4	60.5	7.5
FC-diff [27]	57.3	5.0	56.6	5.0	56.5	5.0
DASNet [59]	56.6	6.2	54.5	5.6	54.0	5.4
HRSCD.str1 [3]	29.3	4.7	30.1	4.9	30.2	5.0
HRSCD.str2 [3]	59.7	5.7	59.0	5.8	59.0	5.9
HRSCD.str3 [3]	62.1	8.4	61.9	8.7	61.9	8.8
HRSCD.str4 [3]	67.2	13.0	67.8	13.8	68.0	14.1
ASN (Ours)	68.4	14.3	68.8	15.2	69.0	15.5
ASN-ATL (Ours)	**69.0**	**14.9**	**69.6**	**15.9**	**69.9**	**16.3**

TABLE IV

COMPARISONS WITH STATE-OF-THE-ART METHODS WHEN THE ENCODER IS BUILT ON SQUEEZE-AND-EXCITATION BLOCKS

Methods	MS __ Flip __		MS ✓ Flip __		MS ✓ Flip ✓	
	mIOU	SeK	mIOU	SeK	mIOU	SeK
FC-EF [27]	60.5	6.9	60.1	6.9	60.1	6.9
FC-conc [27]	64.2	10.5	63.7	10.5	63.6	10.5
FC-diff [27]	63.1	9.7	62.3	9.6	62.3	9.7
DASNet [59]	48.1	-3.5	50.8	-3.8	50.8	-3.7
HRSCD.str1 [3]	30.4	5.0	31.0	5.3	31.1	5.3
HRSCD.str2 [3]	62.3	8.3	61.9	8.3	61.9	8.3
HRSCD.str3 [3]	62.0	8.6	62.0	9.0	62.0	9.1
HRSCD.str4 [3]	67.9	14.6	68.1	15.3	68.2	15.4
ASN (Ours)	69.4	15.9	69.6	16.5	69.7	16.6
ASN-ATL (Ours)	**69.5**	**16.3**	**70.1**	**17.2**	**70.2**	**17.3**

in the SECOND), which can also be seen in Figs. 8–10 (Bottom-Left) and Fig. 11.

We then substitute all the basic residual blocks in the encoder with Xception blocks and squeeze-and-excitation blocks. As illustrated in Tables III and IV, our proposed ASN still outperforms other models under all kinds of testing processes. Especially, under full testing strategies,

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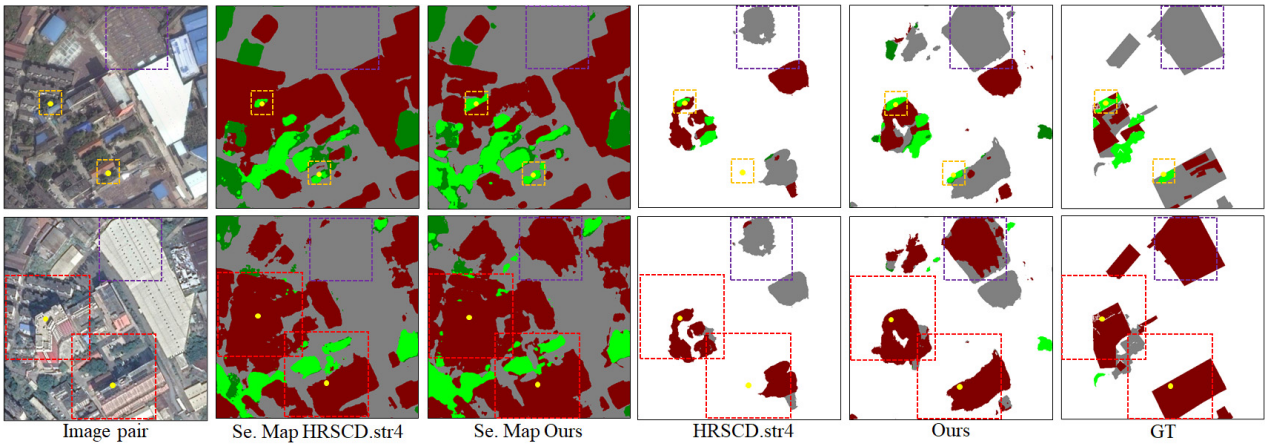


Fig. 11. Results visualization of the case discussed in Section I, including the semantic map (Se.Map) and change map, when the encoder is built on residual blocks. ASN can extract changed regions and identify change types more precisely while addressing these asymmetric changes.

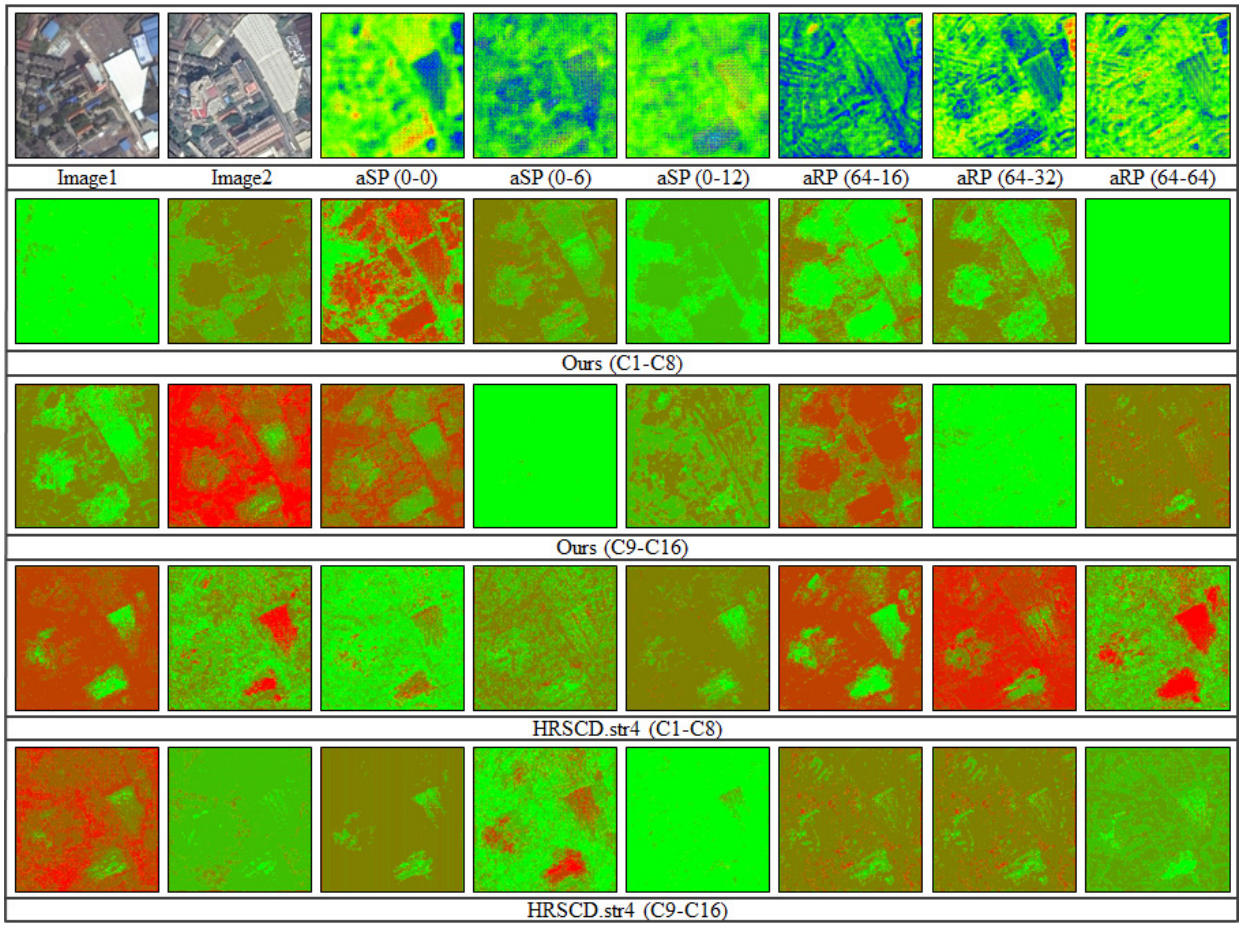


Fig. 12. Feature visualizations through a continuous preserving color mapping function in ASN and HRSCD.str4 with residual blocks. The first row shows the mean value of normalized feature samples w.r.t. different integrations of feature pairs in aSP and aRP. Features deriving from some asymmetric feature pairs, such as aRP (64-32) in the top right, contain more consistent values in regions of asymmetric changes. The rest rows show all 16 channels of corresponding features in the change detection branch of ASN and HRSCD.str4. C1–C8 mean the first eight channels, while C9–C16 mean the last eight channels. Similarly, features in ASN contain more consistent values in regions of asymmetric changes compared with HRSCD.str4.

ASN-ATL outperforms HRSCD.str4 by 1.9 in mIOU and 2.2 in SeK with the Xception blocks, while HRSCD.str4 is inferior to ASN-ATL by 2.0 in mIOU and 1.9 in SeK with squeeze-and-excitation blocks. Meanwhile, ATL provides 0.9 improvements in mIOU and 0.8 improvements in SeK with Xception blocks, while 0.5 improvements in mIOU and 0.7 improvements in SeK can be seen with squeeze-and-excitation blocks. We can conclude that ASN keeps the

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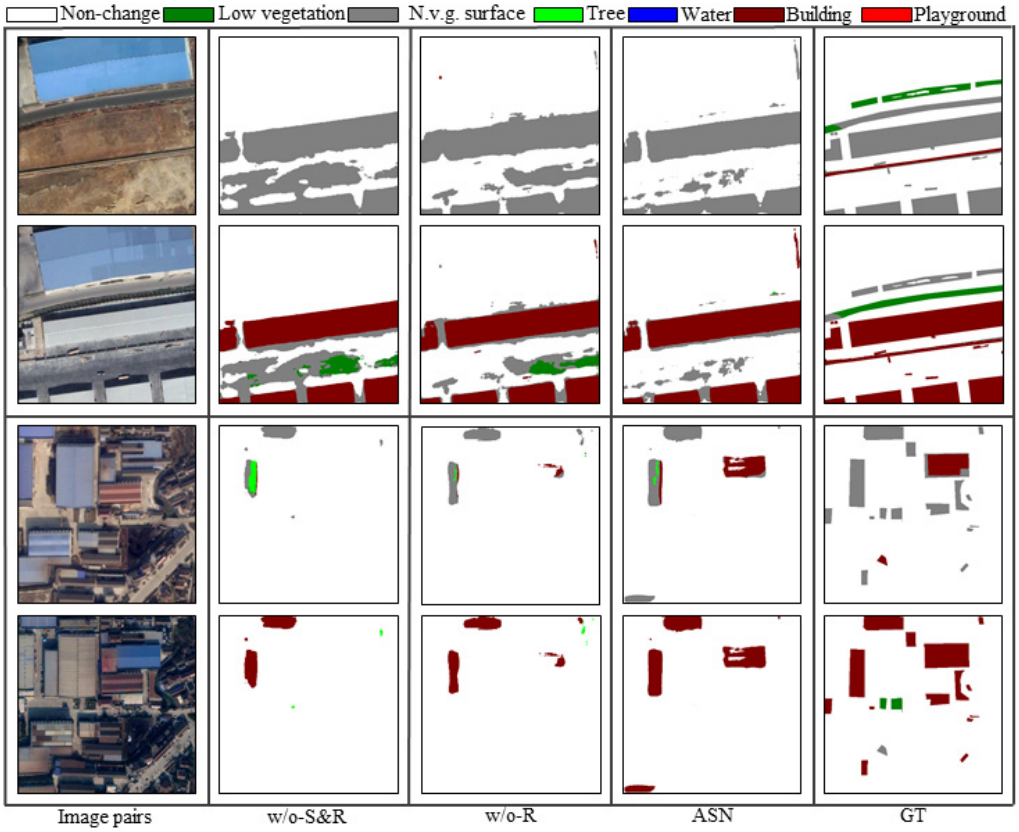


Fig. 13. Results of ablation study with residual blocks. With our proposed modules, the model can better extract obscure change types and alleviate the false identifications of changed pixels.

outperformances with all kinds of backbones, and ATL does improve the model performances. Moreover, Figs. 8–10 report the visual results of HRSCD.str4 and ASN with all three utilized basic blocks. We list the results of the same data samples in the left parts of these figures, which shows the stable effectiveness of ASN with different encoder backbones. Also, we list the results of different data samples in the right parts to exhibit the generalization abilities of the proposed modules.

D. Visualization of Learned Features

Recalling the discussion in Section I, we can see in Fig. 11 that ASN can alleviate the categorical ambiguity caused by asymmetric changes (framed out in colors), where the obscure land-cover categories can be better depicted in obtained semantic maps. We show the visualization of feature map samples in Fig. 12 and distinguish the integration of asymmetric feature pairs by a pair of numbers for each [e.g., aRP (32-64)], where numbers for aSP and aRP indicate the dilation rates and channel numbers of the convolution sequences, respectively. As shown in Fig. 12, the integration of asymmetric feature pairs [e.g., aRP (64-32)], which implicates high representation capabilities w.r.t. former image acquisition time and medium representation capabilities w.r.t. latter image acquisition time, contains more consistent values in regions of asymmetric changes (framed out with colors in Fig. 11) compared with feature pairs deriving from same operations [e.g., aRP (64-64)]. Consequently, as shown in the rest rows of Fig. 12, feature maps [e.g., Ours (C1–C8)] of the decoder for change detection in ASN, which are calculated from the feature pairs in aSP and aRP, contain more consistent values in changed regions compared with HRSCD.str4. Such property of ASN is meaningful for change detection due to the continuity of deep learning networks. In conclusion, asymmetric feature pairs in proposed modules can provide extra information for change detection.

E. Ablation Study

In order to further explore the validity of the proposed modules, we remove aRP and aSP successively to set up the ablation study and check corresponding influences on model performance. As illustrated in Table V, aSP leads to 0.2 improvements in mIOU and 0.6 improvements in SeK with the residual blocks. Then, aRP further improves the model performance by 0.8 in mIOU and SeK. Besides, with the Xception block, aSP improves mIOU by 0.5 and SeK by 0.8, while aRP further improves mIOU by 0.2 and SeK by 0.1. Similarly, with the squeeze-and-excitation block, aSP brings about 0.7 improvements in mIOU and 0.8 improvements in SeK, while aRP raises 0.5 improvements in mIOU and 0.8 improvements in SeK. In summary, aSP and aRP improve model performances on BCD and SCD simultaneously. The effectiveness of our proposed modules can be verified. Moreover, Fig. 13 shows the visual results of the ablation study,

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TABLE V

ABLATION STUDIES OF ASN. W/O-R&S REPRESENTS WITHOUT ASP AND ARP, WHILE W/O-R REPRESENTS WITHOUT ARP

Ablation study	Residual			Xception			SENet		
	w/o-R&S	w/o-R	ASN	w/o-R&S	w/o-R	ASN	w/o-R&S	w/o-R	ASN
mIOU	68.7	68.9	**69.7**	68.3	68.8	**69.0**	68.5	69.2	**69.7**
SeK	14.8	15.4	**16.2**	14.6	15.4	**15.5**	15.0	15.8	**16.6**

TABLE VI

COMPUTATION COSTS CAUSED BY SQUEEZE GATES (GATE) AND CONVOLUTION OPERATIONS (CONV) IN SEQUENCE OF ASP AND ARP. THE NUMBERS FOLLOWING CONV IN ARP SHARE THE SAME MEANING WITH NUMBERS IN FIG. 12. + AND − REPRESENT INCREASING AND DECREASING, RESPECTIVELY

Items	aSP		aRP			
	CONV	GATE	CONV-16	CONV-32	CONV-64	GATE
FLOPS(G)	+ 67.7	- 50.3	+ 6.4	+ 20.9	+ 75.3	- 6.5

TABLE VII

ABLATION STUDIES ON ASYMMETRIC FEATURE PAIRS WITH RESIDUAL BACKBONE. ASN-W/O-AF REPRESENTS RETAINING MULTISCALE STRUCTURES WITHOUT THE ASYMMETRIC FEATURE PAIR INTEGRATION

Methods	MS __ Flip __		MS ✓ Flip __		MS ✓ Flip ✓	
	mIOU	SeK	mIOU	SeK	mIOU	SeK
Single CD	67.7	-	67.8	-	67.8	-
HRSCD.str4 [3]	67.5	13.7	67.8	14.4	67.9	14.5
ASN-w/o-AF	67.7	13.5	68.2	14.4	68.3	14.5
ASN	**69.0**	**15.2**	**69.5**	**16.1**	**69.7**	**16.2**

where we can see that our proposed modules can better extract obscure changed regions, such as the changes between buildings across multitemporal images. In the meantime, the false positive identification of changed regions could be reduced with the proposed modules.

However, the usage of asymmetric feature pairs would cause extra computation costs in ASN. In order to make clear analyzes, we list the computation costs caused by designed squeeze gates and convolution sequences of aSP and aRP in Table VI under the configuration described in Section VI-A, from which we can choose the designed items to generate feature maps for composing the asymmetric feature pairs. In terms of aSP, due to the irrelevance between FLOPS and dilation rates, convolution sequences used to generate asymmetric feature pairs of different receptive fields all cause extra 67.7-G FLOPS, while the utilized squeeze gates have helped to reduce 50.3-G FLOPS by controlling the feature channel numbers and reducing the model sizes. Then, for aRP, convolution sequences of different channel numbers cause extra computation costs to different degrees, while the squeeze gates also help to reduce the FLOPS. Importantly, as we can conclude from Table VI, the feature pair aRP (64-64) shown in Fig. 12 would cause larger extra computation costs, while the asymmetric feature pair aRP (64-32) is still more meaningful for change detection. Although ASN contains the extra cost of computations, the light version of ASN, e.g., ASN-w/o-R, can give the consideration to both effectiveness and computation cost. Moreover, with more concise feature fusion processes of the convolution sequences, the FLOPS of our proposed modules has the potential to be further reduced.

Further to explore the effects of asymmetric feature pairs on binary change detection (BCD) and SCD, we remove the integration of asymmetric feature pairs and whole semantic branches successively. As we can see in Table VII, with residual backbones, the model without integrated asymmetric feature pairs suffers an evident decline in SCD results, which verifies the merits of asymmetric feature pairs on SCD. Then, further without semantic branches, the model with a single change detection branch retains competitive BCD results with HRSCD.str4, whose change detection branch contains extra skip connections from semantic branches. Consequently, we can conclude that simply applying semantic branches on change detection branches through skip connections would not evidently benefit the BCD results. However, by further fusing the asymmetric feature pairs to change detection branches from semantic branches, the intact ASN improves the BCD results significantly.

VII. CONCLUSION

In this article, we propose an ASN for SCD to alleviate categorical ambiguity caused by asymmetric changes through locally asymmetric architectures. To better train deep models, we create a large scale well-annotated SECOND as a new benchmark, which includes the changed regions between the same land-cover class. Furthermore, to alleviate the influence of label imbalance during model training and evaluation, we design an ATL module and an SeK to adaptively revise the output deflections and fix unreasonable scores computed with traditional metrics, respectively. The experimental results show that the proposed model stably achieves the best results with different encoder backbones compared with state-of-the-art algorithms.

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Kunping Yang received the B.S. and M.S. degrees in mathematics from Wuhan University, Wuhan, China, in 2014 and 2016, respectively, where he is currently pursuing the Ph.D. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS).

His research interests include mathematical modeling of images, remote sensing image understanding, and semantic segmentation and change detection.



Gui-Song Xia (Senior Member, IEEE) received the Ph.D. degree in image processing and computer vision from CNRS LTCI, Télécom ParisTech, Paris, France, in 2011.

From 2011 to 2012, he had been a Post-Doctoral Researcher with the Centre de Recherche en Mathématiques de la Decision, CNRS, Paris-Dauphine University, Paris, for one and a half years. He had also been worked as a Visiting Scholar with the DMA, École Normale Supérieure (ENS), Paris, in 2018. He is currently working as a Full Professor of computer vision and photogrammetry jointly with the School of Computer Science, Wuhan University, Wuhan, China, and the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, where he is leading a Research Team named Computational and Photogrammetric Vision Team (CAPTAIN), working toward developing mathematical and computational models to measure and understand our physic environments with vision information. His current research interests include mathematical modeling of images and videos, structure from motion, perceptual grouping, and remote sensing image interpretation.

Dr. Xia also serves on the Editorial Board for the journals *Pattern Recognition*, *Signal Processing: Image Communications*, *EURASIP Journal on Image and Video Processing*, and *Journal of Remote Sensing*. He has served as a Guest Editor for journals, including IEEE TRANSACTIONS ON BIG DATA and *Pattern Recognition Letters*.



Zicheng Liu received the B.S. degree in electrical engineering from Wuhan University of Science and Technology, Wuhan, China, in 2018, and the M.S. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, in 2021.

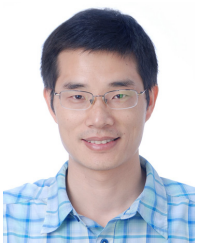
His research interests include mathematical modeling of images, remote sensing image understanding, and semantic segmentation and change detection.



Bo Du (Senior Member, IEEE) received the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, in 2010.

He is currently a Professor with the National Engineering Research Center for Multimedia Software, Wuhan University, the Institute of Artificial Intelligence, Wuhan University, and the School of Computer Science, Wuhan University. He has more than 60 research articles published in the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEM (TNNLS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE JOURNAL OF SELECTED TOPICS IN EARTH OBSERVATIONS AND APPLIED REMOTE SENSING (JSTARS), and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). Thirteen of them are ESI hot articles or highly cited ones. His research interests include pattern recognition, hyperspectral image processing, machine learning, and signal processing.

Dr. Du received the Distinguished Paper Award from IJCAI 2018, the Best Paper Award of IEEE Whispers 2018, the Champion Award of the IEEE Data Fusion Contest 2018, the Highly Cited Researcher 2019 Award from Web of Science Group, and the 2020 Best Paper Award from IEEE TGRS. He also received the ACM Rising Star Award, Wuhan, in 2015. He was the Session Chair for both International Geoscience and Remote Sensing Symposium (IGARSS) 2016/2018 and the 4th IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing. He serves as a Senior PC/PC for IJCAI/AAAI/KDD and an Area Chair for ICPR and IJCNN. He also serves as an Associate Editor for *Pattern Recognition* and *Neurocomputing*.



Wen Yang (Senior Member, IEEE) received the B.S. degree in electronic apparatus and surveying technology, the M.S. degree in computer application technology, and the Ph.D. degree in communication and information system from Wuhan University, Wuhan, China, in 1998, 2001, and 2004, respectively.

From 2008 to 2009, he worked as a Visiting Scholar with the Apprentissage et Interfaces (AI) Team, Laboratoire Jean Kuntzmann (LJK), Grenoble, France. From 2010 to 2013, he worked as a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. Since then, he has been a Full Professor with the School of Electronic Information, Wuhan University. His research interests include object detection and recognition, semantic segmentation, and multisensor information fusion.

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Marcello Pelillo (Fellow, IEEE) has been the Director of the European Centre for Living Technology (ECLT) and has held visiting research/teaching positions in several institutions, including Yale University, New Haven, CT, USA, University College London, London, U.K., McGill University, Montreal, QC, Canada, the University of Vienna, Vienna, Austria, York University, York, U.K., NICTA, Sydney, NSW, Australia, Wuhan University, Wuhan, China, Huazhong University of Science and Technology, Wuhan, and South China University of Technology, Guangzhou, China. He is currently a Full Professor of computer science with Ca' Foscari University of Venice, Venice, Italy, where he leads the Computer Vision and Pattern Recognition Laboratory. He is also an External Affiliate of the Computer Science Department, Drexel University, Philadelphia, PA, USA. His research interests include computer vision, machine learning, and pattern recognition, where he has published more than 200 technical articles in refereed journals, handbooks, and conference proceedings.

Dr. Pelillo has been elected as a fellow of IAPR. He is also an IEEE SMC Distinguished Lecturer. He has been the General Chair of ICCV 2017, the Program Chair of ICPR 2020, and has been the Track Chair or an Area Chair of several conferences in his area. He is also the Specialty Chief Editor of *Frontiers in Computer Vision* and serves, or has served, on the Editorial Board of several journals, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, *Pattern Recognition*, *IET Computer Vision*, and *Brain Informatics*. He also serves on the Advisory Board of Springer's *International Journal of Machine Learning and Cybernetics*. His Erdös number is 2.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He was a Principal Scientist with the China State Key Basic Research Project from 2011 to 2016 appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He is currently the Chair Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He has authored more than 700 research articles and five books. He is the Institute for Scientific Information (ISI) highly cited author. He holds 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is also a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He is also the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter. He also serves as an Associate Editor or an Editor for more than ten international journals. He is also serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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